

TECHNICAL PRINCIPLES	5.3 Why is it important to consider the properties/characteristics of materials when designing and manufacturing products?	
	a. Understand why the characteristics and properties of the materials in 5.2a make them suitable for use in a variety of products dependent on the contextual application, including: density, strength, hardness, durability, strength-to-weight ratio, stiffness, elasticity, impact resistance, plasticity, malleability and ductility, corrosive resistance to chemicals and weather, flammability, absorbency, washability, thermal and electrical conductivity, resistance to decay, biodegradable.	
	b. Understand how the available forms, costs and properties of materials contribute to the decisions about suitability of materials when developing and manufacturing their own products.	

6. Technical understanding

Considerations

*Maths &
Science*